Chapter 11. Groups

Subgroups, homomorphisms, cyclic and linear groups

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11.1 Binary operations and groups

Motivation. Vector addition, matrix multiplication and composition of rigid motions share associativity, identity and inverses: a group.

$X \times Y = \{(x, y) : x \in X, y \in Y\}$ is the **Cartesian product**; (x, y) is an **ordered pair**.

Definition 11.1.1 (Binary operation)

A function $* : X \times X \to X$ is a **binary operation** on X; we write $x * y$, or $x \cdot y$, xy.

Definition 11.1.2 (Associativity)

A binary operation is **associative** if $(xy)z = x(yz)$ for all $x, y, z \in X$; then xyz is meaningful.

Example 11.1.3

On $\mathfrak{M}_{n,n}(F)$, the bracket $[A, B] = AB - BA$ is not associative: for $n = 2$, $[[E_{12}, E_{21}], E_{21}] = [E_{11} - E_{22}, E_{21}] = -2E_{21}$, while $[E_{12}, [E_{21}, E_{21}]] = 0$.

From now on every binary operation is assumed associative.

Definition 11.1.4 (Group)

A set G with a binary operation is a **group** if (G1) $(xy)z = x(yz)$ for all $x, y, z \in G$; (G2) there is $e \in G$ with $xe = x = ex$ for all $x \in G$; (G3) for each $x \in G$ there is $\tilde{x} \in G$ with $x\tilde{x} = e = \tilde{x}x$.

Example 11.1.5

(a) $(\{-1, 1\}, +)$ is not a group ($1 + 1 \notin \{-1, 1\}$), but $(\{-1, 1\}, \times)$ is.
(b) $\{e\}$ with $ee = e$ is the **trivial group**.
(c) $(\mathbb{Z}, +)$ is a group; $(\mathbb{Z}, \times)$ is not.
(d) $(F, +)$ and $(F - \{0\}, \times)$ are groups; $(F, \times)$ is not.
(e) For a vector space V, $(V, +)$ is a group (axioms (V1)–(V4)).
(f) The symmetric group $(S_n, \circ)$ of Chapter 6 is a group.

Observation 11.1.6

A group is not empty.

Proof. By (G2) it contains e. ■

Observation 11.1.7

The element e of (G2) is unique; it is the **identity** of G, written e_G.

Proof. If e' is another, $e = ee' = e'$. ■

Observation 11.1.8

For each $x \in G$, the $\tilde{x}$ of (G3) is unique; it is the **inverse** x^{-1}.

Proof. If $xx' = e = x'x$ as well, then $\tilde{x} = \tilde{x}e = \tilde{x}(xx') = (\tilde{x}x)x' = ex' = x'$. ■

Exercise 11.1.9

For $g, h \in G$: (a) $(gh)^{-1} = h^{-1}g^{-1}$;
(b) $(g^{-1})^{-1} = g$.

Solution. (a) $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = e$ and likewise on the other side; use uniqueness of inverses.

(b) g satisfies the defining property of the inverse of g^{-1}. □

Definition 11.1.10 (Order)

G is a **finite group** if it is a finite set; then $|G|$ is its **order**. Otherwise G is an **infinite group**.

Definition 11.1.11 (Commutative group)

G is **commutative** (**abelian**) if $xy = yx$ for all $x, y \in G$.

Exercise 11.1.12

$\mu_n = \{\alpha \in \mathbb{C} : \alpha^n = 1\}$, under multiplication of complex numbers, is a finite commutative group of order n.

Solution. μ_n contains 1 and is closed under products and inverses; associativity and commutativity come from $\mathbb{C}$. Its elements are the n distinct $e^{2\pi ik/n}$, $0 \le k < n$ (at most n roots of $t^n - 1$). □

Exercise 11.1.13

For $n \ge 3$, S_n is a finite non-commutative group of order $n!$.

Solution. $|S_n| = n!$. $(1\,2)(1\,3)$ sends $1 \mapsto 3 \mapsto 3$ but $(1\,3)(1\,2)$ sends $1 \mapsto 2$; so they differ. □

Exercise 11.1.14

On F^3 define $(a_1, a_2, a_3)^t \bullet (b_1, b_2, b_3)^t = (a_1 + b_1,\ a_2 + b_2 + a_1b_3,\ a_3 + b_3)^t$. Then $(F^3, \bullet)$ is an infinite non-commutative group.

Solution. Both $(a\bullet b)\bullet c$ and $a\bullet(b\bullet c)$ equal $(a_1+b_1+c_1,\ a_2+b_2+c_2+a_1b_3+a_1c_3+b_1c_3,\ a_3+b_3+c_3)$. The identity is 0, and the inverse of a is $(-a_1, -a_2+a_1a_3, -a_3)$. Non-commutative: $e_1 \bullet e_3 = (1, 1, 1)$ but $e_3 \bullet e_1 = (1, 0, 1)$. (This is the Heisenberg group of upper unitriangular 3×3 matrices in disguise.) □

11.2 Elementary properties of groups

Motivation. Powers, cancellation, and multiplication tables classify the smallest groups.

Notation 11.2.1

(a) (Multiplicative notation) The identity is 1; $g^1 = g$, $g^2 = gg$, ..., $g^0 = 1$, $g^{-2} = g^{-1}g^{-1}$, ...
(b) (Additive notation) For an abelian group A written with $+$: identity 0, inverse $-a$, and $\ldots, (-2)a = (-a) + (-a)$, $0a = 0$, $1a = a$, $2a = a + a, \ldots$; $a - b = a + (-b)$.
(c) Commutative groups are usually written multiplicatively and abelian groups additively.
(d) Unless stated otherwise we use multiplicative notation.

Exercise 11.2.2

(a) $(g^n)^{-1} = (g^{-1})^n = g^{-n}$ for $n \in \mathbb{Z}$.
(b) Additively: $-(na) = n(-a) = (-n)a$.

Solution. $g^n(g^{-1})^n = 1$ by cancelling adjacent gg^{-1}; and $g^{-n} = (g^{-1})^n$ by definition (for $n > 0$; the other cases follow). □

Exercise 11.2.3

$(h^{-1}gh)^n = h^{-1}g^nh$ for $n \in \mathbb{Z}$.

Solution. For $n > 0$, the inner hh^{-1} cancel; $n = 0$ is clear; for $n < 0$ apply this to g^{-1}, using $(h^{-1}gh)^{-1} = h^{-1}g^{-1}h$. □

Exercise 11.2.4

Elements $ghg^{-1}h^{-1}$ are **commutators**. G is commutative iff all commutators are 1.

Solution. $ghg^{-1}h^{-1} = 1 \iff gh = hg$. □

Exercise 11.2.5

If $g^2 = 1$ for all $g \in G$, then G is commutative.

Solution. Then $g = g^{-1}$ for all g, so $ghg^{-1}h^{-1} = ghgh = (gh)^2 = 1$. □

Exercise 11.2.6 (Exponent laws, multiplicative)

For $g, h \in G$, $m, n \in \mathbb{Z}$: (a) $g^m g^n = g^{m+n}$;
(b) $(g^m)^n = g^{mn}$;
(c) if G is commutative, $g^n h^n = (gh)^n$.

Solution. For $m, n \geq 0$ count factors; negative exponents reduce to this via $g^{-k} = (g^{-1})^k$. (c): regroup $(gh)\cdots(gh)$ by commutativity. □

Exercise 11.2.7 (Exponent laws, additive)

For $a, b \in A$, $m, n \in \mathbb{Z}$: (a) $ma + na = (m + n)a$;
(b) $m(na) = (mn)a$;
(c) $na + nb = n(a + b)$.

Solution. Exercise 11.2.6 in additive notation. □

Observation 11.2.8 (Cancellation law)

If $xy = xz$, then $y = z$; if $xz = yz$, then $x = y$.

Proof. $y = (x^{-1}x)y = x^{-1}(xy) = x^{-1}(xz) = z$; the other side is symmetric. ■

Exercise 11.2.9

State Observation 11.2.8 additively.

Solution. If $a + b = a + c$ then $b = c$; if $a + c = b + c$ then $a = b$. □

Example 11.2.10

$xy = x$ implies $y = 1$, since $xy = x1$.

Exercise 11.2.11

(a) If $y = u^{-1}xu$, then $x = uyu^{-1}$.
(b) If $u^{-1}xu = 1$, then $x = 1$.

Solution. (a) Multiply by u on the left and u^{-1} on the right.

(b) By (a), $x = u1u^{-1} = 1$. □

A finite group $G = \{1, x_2, x_3, \dots\}$ can be described by its **multiplication table**, whose (x_i, x_j) entry is $x_i x_j$.

Example 11.2.12

All groups of order 1 are "the same": $\{e\}$ and $(\{0\}, +)$ differ only in names.

Example 11.2.13

If $G = \{1, x\}$, then $x^2 \neq x$ (cancellation), so $x^2 = 1$: the table is forced, and G is μ_2 up to renaming.

Observation 11.2.14

In a multiplication table every row contains each element exactly once, and so does every column.

Proof. $xy = xz$ forces $y = z$ (cancellation), so a row has no repetitions; G is finite, so each element appears. Similarly for columns. ■

Observation 11.2.15

For fixed $x \in G$, $\lambda_x : G \to G$, $\lambda_x(y) = xy$, is a bijection (it permutes G).

Proof. Injective by cancellation; surjective since $z = \lambda_x(x^{-1}z)$. ■

Example 11.2.16

If $G = \{1, x, y\}$, then $x^2 \neq x$, and $x^2 = 1$ would force $xy = y$, contradicting cancellation; so $x^2 = y$, and by Observation 11.2.14 the table is

	1	x	y
1	1	x	y
x	x	y	1
y	y	1	x

So there is at most one group of order 3 up to renaming.

Question 11.2.17

How can we check that a given table is associative?

If a known group realises the table, associativity is automatic: for $\zeta = e^{2\pi i/3}$, the renaming $x \mapsto \zeta$, $y \mapsto \zeta^2$ carries the table of Example 11.2.16 to that of μ_3.

Example 11.2.18

For $G = \{1, x, y, z\}$, cancellation and Observation 11.2.14 leave four possible tables (up to symmetry): one in which every element squares to 1, and three in which one non-identity element squares to 1 (the second, third and fourth tables), which are $\mu_4 = \{1, -1, i, -i\}$ up to renaming.

Exercise 11.2.19

The second, third and fourth tables of Example 11.2.18 are associative.

Solution. Each is the table of μ_4 after a renaming (e.g. $x \mapsto -1$, $y \mapsto i$, $z \mapsto -i$). □

Example 11.2.20

For groups G_1, G_2, the Cartesian product $G_1 \times G_2$ with $(g_1, g_2)(h_1, h_2) = (g_1h_1, g_2h_2)$ is a group, the **direct product**.

Example 11.2.21

$\mu_2 \times \mu_2$ realises the first table of Example 11.2.18; e.g. $(1, -1)(1, -1) = (1, 1)$. It is the **Klein 4-group**, not μ_4 (all its elements square to 1).

Exercise 11.2.22

(If you can,) show with tables that a group of order 5 is μ_5 up to renaming.

Solution [Supplement] (via Lagrange's theorem, Observation 12.1.10). Let $x \neq 1$. The powers of x form a subgroup $\langle x \rangle$ (§11.3) whose order divides 5, hence equals 5. So $G = \{1, x, x^2, x^3, x^4\}$ with $x^5 = 1$, and $x^k \mapsto e^{2\pi ik/5}$ is a renaming onto μ_5 preserving the table. □

Exercise 11.2.23

If $|G| \leq 5$, then G is commutative; so S_3 is the smallest non-commutative group.

Solution. By Examples 11.2.12, 11.2.13, 11.2.16, 11.2.18 and Exercise 11.2.22, every such group is μ_n or $\mu_2 \times \mu_2$, whose tables are symmetric. S_3 has order 6 and is non-commutative. □

Remark 11.2.24

Semigroups, monoids, and one-sided identities and inverses appear in other texts; they are not needed here, nor is the classification of finite groups.

11.3 Subgroups

Motivation. Subgroups play the role of subspaces; the subgroup generated by S consists of words in $S \cup S^{-1}$.

Definition 11.3.1 (Subgroup)

A subset $H \subseteq G$ that is a group under the operation of G is a **subgroup**, $H \leq G$.

Observation 11.3.2

Let $H \leq G$. (a) $e_H = e_G$.
(b) The inverse of $h \in H$ in H equals its inverse h^{-1} in G.

Proof. (a) In G, $e_H e_H = e_H = e_H e_G$; cancel e_H.

(b) In G, $hh' = e_H = e_G = hh^{-1}$; cancel h. ■

Exercise 11.3.3

Why was $0_W = 0_V$ for subspaces $W \leq V$ never an issue?

Solution. $0_W = 0 \cdot w = 0_V$ for any $w \in W$. □

Observation 11.3.4

$H \subseteq G$ is a subgroup iff (S1) $h_1, h_2 \in H \Rightarrow h_1 h_2 \in H$; (S2) $e \in H$; (S3) $h \in H \Rightarrow h^{-1} \in H$.

Proof. Necessity: the operation must restrict to H (S1), and Observation 11.3.2 gives (S2), (S3). Sufficiency: associativity is inherited from G. ■

Exercise 11.3.5

A nonempty $H \subseteq G$ is a subgroup iff $h_1, h_2 \in H \Rightarrow h_1 h_2^{-1} \in H$.

Solution. Necessity is clear. For sufficiency, take $h \in H$: then $e = hh^{-1} \in H$, $h^{-1} = eh^{-1} \in H$, and $h_1 h_2 = h_1 (h_2^{-1})^{-1} \in H$. □

Example 11.3.6

(a) $\{e\} \leq G \leq G$. (b) Under addition, $\mathbb{Z} < \mathbb{Q} < \mathbb{R} < \mathbb{C}$. (c) Under multiplication, $\mu_2 < \mathbb{Q}-\{0\} < \mathbb{R}-\{0\} < \mathbb{C}-\{0\}$. (d) The alternating group A_n is a subgroup of S_n. (e) If $m \mid n$, then $\mu_m \leq \mu_n$. (f) If W is a subspace of V, then $(W,+) \leq (V,+)$.

Observation 11.3.7

If $K \leq H$ and $H \leq G$, then $K \leq G$.

Proof. K is a group under the operation of H, which is that of G. ■

Exercise 11.3.8

If $I \neq \emptyset$ and $H_i \leq G$ for all $i \in I$, then $\bigcap_i H_i \leq G$.

Solution. Each H_i contains e and is closed under products and inverses; so is the intersection. □

Notation 11.3.9

For $H, K \subseteq G$, $HK = \{hk : h \in H, k \in K\}$.

Exercise 11.3.10

(a) If G is commutative and $H, K \leq G$, then $HK \leq G$.
(b) In $S_3 = \{id, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$, $H = \{id, (1\,2)\}$ and $K = \{id, (2\,3)\}$ are subgroups but $HK \not\leq S_3$.

Solution. (a) $(h_1k_1)(h_2k_2)^{-1} = h_1h_2^{-1}k_1k_2^{-1} \in HK$ by commutativity; apply Exercise 11.3.5. (b) $HK = \{id, (2\,3), (1\,2), (1\,2)(2\,3)\}$ does not contain $((1\,2)(2\,3))^{-1} = (2\,3)(1\,2)$. □

Definition 11.3.11 (Generated subgroup)

For $S \subseteq G$, the smallest subgroup of G containing S is the **subgroup generated by** S, $\langle S \rangle$: $S \subseteq \langle S \rangle \leq G$, and $S \subseteq H \leq G$ implies $\langle S \rangle \subseteq H$.

Observation 11.3.12

$\langle S \rangle = \bigcap_{S \subseteq H \leq G} H$; in particular $\langle S \rangle$ exists and is unique.

Proof. By Exercise 11.3.8 the intersection is a subgroup containing S, and it lies in every subgroup containing S. ■

Example 11.3.13

$\langle \emptyset \rangle = \{e\}$, and $\langle x \rangle = \{x^i : i \in \mathbb{Z}\}$ (a subgroup by the exponent laws); possibly $x^i = x^j$ for $i \neq j$.

Definition 11.3.14 (Cyclic group)

(a) $\langle x \rangle$ is the **cyclic subgroup** generated by x.
(b) If $G = \langle x \rangle$ for some x, G is a **cyclic group** with generator x. Cyclic groups are commutative (§11.7).

Definition 11.3.15 (Alphabet and words)

(a) For $S \subseteq G$ let $S^{-1} = \{s^{-1} : s \in S\}$; $S \cup S^{-1}$ is the **alphabet**.
(b) Finite products of letters are **words**; e is the empty word. (Different words may represent the same element.)

Observation 11.3.16

The set of all words in the alphabet $S \cup S^{-1}$ is a subgroup of G.

Proof. It contains the empty word, products of words are words, and $(s_1 \cdots s_k)^{-1} = s_k^{-1} \cdots s_1^{-1}$ is a word (Exercise 11.1.9); apply Observation 11.3.4. ■

Proposition 11.3.17

$\langle S \rangle$ is the set K of all words in the alphabet $S \cup S^{-1}$.

Proof. $\langle S \rangle$ is a subgroup containing S, so it contains all such words: $K \subseteq \langle S \rangle$. Conversely K is a subgroup (Observation 11.3.16) containing S, so $\langle S \rangle \subseteq K$. ■

Exercise 11.3.18

(a) $(\mathbb{Z}, +)$ is cyclic; how many generators does it have?
(b) $3\mathbb{Z} = \{3a : a \in \mathbb{Z}\}$ is a cyclic subgroup of $(\mathbb{Z}, +)$.
(c) $\langle 2, 3 \rangle = \mathbb{Z} = \langle 4, 15 \rangle$, but $\langle 6, 9 \rangle \neq \mathbb{Z}$.

Solution. (a) $\mathbb{Z} = \langle 1 \rangle = \langle -1 \rangle$, and these are the only generators: $\langle n \rangle = n\mathbb{Z} \neq \mathbb{Z}$ for $|n| \neq 1$.

(b) $3\mathbb{Z} = \langle 3 \rangle$.

(c) $1 = 3 - 2 = 4 \cdot 4 - 15$, so 1 lies in both; but every element of $\langle 6, 9 \rangle$ is a multiple of 3, so $\langle 6, 9 \rangle = 3\mathbb{Z}$. □

Exercise 11.3.19

(a) μ_5 is cyclic; how many generators?
(b) μ_6 is cyclic; how many generators?

Solution. $\mu_n = \langle \zeta \rangle$ with $\zeta = e^{2\pi i/n}$, and ζ^k generates iff $\gcd(k, n) = 1$. (a) 4 generators ($k = 1, 2, 3, 4$).

(b) 2 generators ($k = 1, 5$). □

Exercise 11.3.20

(a) Find all subgroups of S_3.
(b) Is S_3 cyclic?
(c) $\langle (1\,2), (2\,3) \rangle = S_3 = \langle (1\,2), (1\,2\,3) \rangle$.

Solution. (a) $\{id\}$, $\langle(1\,2)\rangle$, $\langle(1\,3)\rangle$, $\langle(2\,3)\rangle$, $A_3 = \langle(1\,2\,3)\rangle$, S_3 (two transpositions, or a transposition and a 3-cycle, generate S_3 by (c)).

(b) No: element orders are $1, 2, 3$.

(c) $(1\,2)(2\,3)$, $(2\,3)(1\,2)$ are the two 3-cycles and $(1\,2)(2\,3)(1\,2) = (1\,3)$. Likewise $(1\,2\,3)^2$, $(1\,2)(1\,2\,3)$, $(1\,2\,3)(1\,2)$ give the rest. □

11.4 The heart of undergraduate algebra

Motivation. Every subgroup of $(\mathbb{Z}, +)$ is $n\mathbb{Z}$, by division by its least positive element.

Example 11.4.1

For $n \in \mathbb{Z}$, $n\mathbb{Z} = \{na : a \in \mathbb{Z}\} = \langle n\rangle$ is a subgroup of $(\mathbb{Z}, +)$.

Theorem 11.4.2 (The heart of undergraduate algebra)

The subgroups of $(\mathbb{Z}, +)$ are exactly the $n\mathbb{Z}$, $0 \le n \in \mathbb{Z}$.

Proof. Let $0 \ne H \le \mathbb{Z}$ and n its least positive element. For $a \in H$, $a = qn + r$ with $0 \le r < n$ gives $r = a - qn \in H$, so $r = 0$. Hence $H \le n\mathbb{Z} = \langle n\rangle \le H$. ($H = 0$ is $0\mathbb{Z}$.) ■

11.5 Group isomorphisms

Motivation. Isomorphic groups differ only in the names of their elements.

Definition 11.5.1 (Isomorphism)

Groups G, G' are **isomorphic**, $G \approx G'$, if there is a bijection $\varphi : G \to G'$ with $\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2)$; φ is a (**group**) **isomorphism**. Equivalently, with $(\varphi \times \varphi)(g_1, g_2) = (\varphi g_1, \varphi g_2)$, the square

$$\begin{array}{ccc} G \times G & \xrightarrow{*} & G \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ G' \times G' & \xrightarrow[*']{} & G' \end{array}$$

commutes.

Example 11.5.2

Let $\varphi : G \to H$ be an isomorphism, $g \in G$, $S \subseteq G$. Then (a) $\varphi(e_G) = e_H$;
(b) $\varphi(g^{-1}) = \varphi(g)^{-1}$;
(c) if $\langle S\rangle = G$, then $\langle\varphi(S)\rangle = H$;
(d) if $g^2 = e_G$, then $\varphi(g)^2 = e_H$;
(e) if G is commutative, so is H;
(f) if G is cyclic, so is H;
(g) if G has six subgroups, so does H.

Proof [Supplement]. (a) $\varphi(e)\varphi(e) = \varphi(e) = \varphi(e)e_H$; cancel.

(b) $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e_H$.

(c) φ maps words in $S \cup S^{-1}$ onto words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17) and is onto.

(d) Apply φ.

(e) $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$, and φ is onto.

(f) Case of (c).

(g) $K \mapsto \varphi(K)$ is a bijection between the sets of subgroups (with inverse $K' \mapsto \varphi^{-1}(K')$). ■

Exercise 11.5.3

Is $\mathbb{Z} \approx \mathbb{Z} \times \mathbb{Z}$ (additive groups)?

Solution. No (Example 11.5.2(f)): $\mathbb{Z} \times \mathbb{Z}$ is not cyclic, as $\langle (a,b) \rangle = \{(na, nb)\}$ never contains both $(1,0)$ and $(0,1)$. □

Example 11.5.4

Properties *not* preserved by isomorphisms: "the operation is called multiplication", "G has an element named x", "the elements are complex numbers", "$|g| = 7$ as a real number", "g integrates to h".

Observation 11.5.5

$\approx$ is an equivalence relation.

Proof. Exercise 11.5.6. ■

Exercise 11.5.6

(a) id_G is an isomorphism.
(b) If $\varphi : G \to H$ is an isomorphism, so is φ^{-1}.
(c) If $\psi : K \to G$, $\varphi : G \to H$ are isomorphisms, so is $\varphi \circ \psi$.

Solution. (b) With $h_i = \varphi(g_i)$: $\varphi^{-1}(h_1 h_2) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$. (a), (c) are direct. □

Example 11.5.7

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a group isomorphism, with inverse log: $e^{a+b} = e^a e^b$.

Remark 11.5.8

Unlike vector spaces (dimension), deciding whether two groups are isomorphic is in general hard.

11.6 Group homomorphisms

Motivation. Homomorphisms are the analogue of linear maps, with kernels and images.

Definition 11.6.1 (Homomorphism)

$\varphi : G \to H$ is a (**group**) **homomorphism** if $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$.

Observation 11.6.2

For a homomorphism φ: (a) $\varphi(e) = e$;
(b) $\varphi(x^{-1}) = \varphi(x)^{-1}$.

Proof. As in Example 11.5.2(a), (b). ■

Definition 11.6.3 (Kinds of homomorphisms)

A homomorphism $\varphi : G \to H$ is a **monomorphism** if injective, an **epimorphism** if surjective, an **isomorphism** if bijective, an **endomorphism** if $G = H$, and an **automorphism** if it is a bijective endomorphism.

Observation 11.6.4

A homomorphism $\varphi : G \to H$ is an isomorphism iff there is a homomorphism $\psi : H \to G$ with $\psi\varphi = id_G$ and $\varphi\psi = id_H$.

Proof. If φ is an isomorphism, take $\psi = \varphi^{-1}$ (Exercise 11.5.6). Conversely such a ψ is an inverse function, so φ is bijective. ■

Definition 11.6.5 (Kernel and image)

$\ker \varphi = \varphi^{-1}(e) = \{x \in G : \varphi(x) = e\}$ and $\operatorname{im} \varphi = \varphi(G)$.

Observation 11.6.6

For a homomorphism $\varphi : G \to H$: (a) $\ker \varphi \leq G$ and $\operatorname{im} \varphi \leq H$;
(b) if φ is a monomorphism, $G \approx \operatorname{im} \varphi$;
(c) if $K \leq G$, then $\varphi(K) \leq H$.

Proof [Supplement]. (a) If $\varphi x = \varphi y = e$, then $\varphi(xy^{-1}) = e$, and $e \in \ker \varphi$ (Exercise 11.3.5).

(c) $\varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(K)$; the case $K = G$ gives $\operatorname{im} \varphi \leq H$.

(b) $\varphi : G \to \operatorname{im} \varphi$ is a bijective homomorphism. ■

Observation 11.6.7

For a homomorphism φ the following are equivalent: (1) φ is a monomorphism; (2) $\varphi x = \varphi y \Rightarrow x = y$; (3) $\varphi x = \varphi e = e \Rightarrow x = e$; (4) $\ker \varphi = \{e\}$.

Proof. All but (3)⇒(2) are immediate. If (3) holds and $\varphi x = \varphi y$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = e$, so $xy^{-1} = e$, $x = y$. ■

Observation 11.6.8

For a homomorphism $\varphi : G \to H$ and $S \subseteq G$: (a) $\varphi\langle S\rangle = \langle \varphi S\rangle$;
(b) if $G = \langle S\rangle$, then $\operatorname{im} \varphi = \langle \varphi S\rangle$;

(c) φ is an epimorphism iff it maps a generating set of G to a generating set of H.

Proof. (a) Images of words in $S \cup S^{-1}$ are words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17), so $\varphi\langle S\rangle \leq \langle \varphi S\rangle$; conversely $\varphi\langle S\rangle \leq H$ contains φS (Observation 11.6.6(c)).

(b) The case $\langle S\rangle = G$.

(c) By (b), $\operatorname{im}\varphi = H$ iff $\langle \varphi S\rangle = H$. ■

Exercise 11.6.9

If $\varphi : G \to H$ is a homomorphism and G is cyclic, then $\operatorname{im}\varphi$ is cyclic.

Solution. If $G = \langle x\rangle$, then $\operatorname{im}\varphi = \langle \varphi x\rangle$ by Observation 11.6.8(b). □

Exercise 11.6.10

Let $\varphi(g) = g^{-1}$ and $\psi(g) = g^2$ on G. (a) φ is a bijection; is it a homomorphism?
(b) Is ψ a homomorphism?

Solution. (a) $\varphi = \varphi^{-1}$; it is a homomorphism iff $y^{-1}x^{-1} = x^{-1}y^{-1}$ for all x, y, iff G is commutative.

(b) $xyxy = xxyy$ iff $yx = xy$: again iff G is commutative. □

Exercise 11.6.11

(a) For $n \geq 2$, $\operatorname{sgn} : S_n \to \mu_2$ is an epimorphism.
(b) Describe $\ker(\operatorname{sgn})$.

Solution. (a) $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \operatorname{sgn}\tau$, and $\operatorname{sgn}(1\,2) = -1$.

(b) $\ker(\operatorname{sgn}) = A_n$, the even permutations. □

11.7 Cyclic groups

Motivation. Up to isomorphism a cyclic group is $\mathbb{Z}$ or μ_n.

Example 11.7.1

(a) $\mathbb{Z} = \langle 1\rangle$ is an infinite cyclic group, and so is $n\mathbb{Z} = \langle n\rangle$ for $n > 0$; $a \mapsto na$ is an isomorphism $\mathbb{Z} \approx n\mathbb{Z}$ onto a proper subgroup.
(b) For $\zeta = e^{2\pi i/n}$, $\mu_n = \{1, \zeta, \ldots, \zeta^{n-1}\} = \langle \zeta\rangle$ is cyclic of order n.

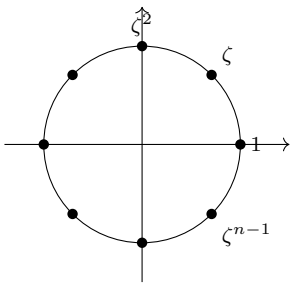


Figure. *μ_n ($n = 8$): the vertices of a regular n-gon on the unit circle.*

Definition 11.7.2 (Order of an element)

The **order** of $x \in G$ is $|x| = |\langle x \rangle|$.

Observation 11.7.3

If x has finite order, then $|x| = n$ iff (1) $x^n = 1$ and (2) $x^m \neq 1$ for $0 < m < n$; i.e. n is the least positive integer with $x^n = 1$.

Proof. $\langle x \rangle$ is finite, so $x^i = x^j$ for some $i < j$: $\{a \geq 1 : x^a = 1\} \neq \emptyset$; let n be its least element (the n of (1), (2)). For $b = qn + r$, $0 \leq r < n$, $x^b = (x^n)^q x^r = x^r$, so $\langle x \rangle = \{1, x, \ldots, x^{n-1}\}$; these are distinct, since $x^k = x^\ell$, $0 \leq k < \ell < n$, gives $x^{\ell-k} = 1$. So $|x| = n$. ■

Definition 11.7.4 (Order, second definition)

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, is a homomorphism (exponent laws), so $\ker \gamma_x = n\mathbb{Z}$ for a unique $n \geq 0$ (Theorem 11.4.2). Set $|x| = \infty$ if $n = 0$ and $|x| = n$ otherwise.

Exercise 11.7.5

Definitions 11.7.2 and 11.7.4 agree.

Solution. If $\ker \gamma_x = 0$, γ_x is injective and $\langle x \rangle = \operatorname{im} \gamma_x$ is infinite. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, then n is the least positive integer with $x^n = 1$, and $|\langle x \rangle| = n$ by Observation 11.7.3. □

Exercise 11.7.6

(a) $|x| = 1$ iff $x = 1$.
(b) $|x| = 2$ iff $1 \neq x = x^{-1}$.
(c) Every nonzero element of $(\mathbb{R}, +)$ has order ∞.
(d) A k-cycle in S_n has order k.

Solution. (a),

(b) by Observation 11.7.3.

(c) $na \neq 0$ for $n \neq 0$, $a \neq 0$.

(d) $(i_1 \ldots i_k)^m$ sends $i_1 \mapsto i_{1+m}$ (indices mod k), which is i_1 iff $k \mid m$; the k-th power is id. □

Exercise 11.7.7

If $|x| = n$ and $a, b \in \mathbb{Z}$, then $x^a = x^b$ iff $n \mid (a - b)$.

Solution. $x^a = x^b \iff x^{a-b} = 1 \iff a - b \in \ker \gamma_x = n\mathbb{Z}$. □

Exercise 11.7.8

Every subgroup of a cyclic group is cyclic.

Solution. Let $G = \langle x \rangle$, $H \leq G$. Then $\gamma_x : \mathbb{Z} \to G$ is onto, $\gamma_x^{-1}(H) \leq \mathbb{Z}$ equals $m\mathbb{Z}$ (Theorem 11.4.2), and $H = \gamma_x(m\mathbb{Z}) = \langle x^m \rangle$. □

Exercise 11.7.9

Up to isomorphism the cyclic groups are $\mathbb{Z}$ and the μ_n.

Solution. Let $G = \langle x \rangle$. If $\ker \gamma_x = 0$, $\gamma_x : \mathbb{Z} \to G$ is an isomorphism. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, $\varphi(x^a) = \zeta^a$ is well defined and injective since $x^a = x^b \iff n \mid a-b \iff \zeta^a = \zeta^b$ (Exercise 11.7.7); it is onto and a homomorphism. □

11.8 Examples of groups and homomorphisms

Motivation. Symmetric groups, Cayley's theorem, conjugacy, automorphisms and products.

Exercise 11.8.1

(a) Composites of homomorphisms are homomorphisms.
(b) If $L \in \mathfrak{L}(V, W)$, then $L : (V, +) \to (W, +)$ is a homomorphism.

Solution. (a) $\varphi\psi(xy) = \varphi(\psi x \psi y) = \varphi\psi x \, \varphi\psi y$.

(b) $L(v + w) = Lv + Lw$. □

Exercise 11.8.2

(a) The constant map $\mathbf{1}(g) = 1$ is a homomorphism, the **trivial map** (written 0 for abelian groups). (b) For $H \leq G$, the inclusion $\iota_H : H \to G$ is a monomorphism, the **natural embedding**. (c) For $H \leq G$ and a homomorphism $\varphi : G \to G'$, the restriction $\varphi|_H$ is a homomorphism.

Solution. All follow directly from the definitions ($1 \cdot 1 = 1$; the operation of H is that of G). □

Example 11.8.3

"Every permutation is a product of transpositions" says S_n is *generated* by transpositions. $\operatorname{sgn} : S_n \to \mu_2$ is a homomorphism with $\ker(\operatorname{sgn}) = A_n$, so $A_n \leq S_n$ automatically (Observation 11.6.6).

Example 11.8.4

(a) In S_3, put $\sigma = (1\,2)$, $\tau = (1\,2\,3)$. Then $\sigma^2 = id = \tau^3$, $\tau^2 = (1\,3\,2)$, $\sigma\tau = (2\,3)$, $\sigma\tau^2 = (1\,3)$, so

$$S_3 = \{id, \tau, \tau^2, \sigma, \sigma\tau, \sigma\tau^2\} = \{\sigma^i\tau^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\} = \langle \sigma, \tau \rangle.$$

(b) $\sigma^{-1}\tau\sigma = \tau^2 = \tau^{-1}$, i.e. $\tau\sigma = \sigma\tau^2$, moves σ to the left, so the relations

$$\sigma^2 = 1, \qquad \tau^3 = 1, \qquad \tau\sigma = \sigma\tau^2$$

determine the whole multiplication table of S_3; e.g. $(\sigma\tau)(\sigma\tau^2) = \sigma(\tau\sigma)\tau^2 = \sigma\sigma\tau^2\tau^2 = \tau$.
(c) This is the plane rule of Observation 9.5.9; see Example 11.9.23.

Definition 11.8.5 (Symmetric group on a set)

For a nonempty set X, $S_X = \{\sigma : X \to X \text{ bijection}\}$ is a group under composition, the **symmetric group** on X.

Exercise 11.8.6

(a) How should "X and Y are the same as sets" be defined?
(b) If X, Y are isomorphic as sets, then $S_X \approx S_Y$.

Solution. (a) There is a bijection $f : X \to Y$ (a renaming).

(b) $\sigma \mapsto f \circ \sigma \circ f^{-1}$ is a bijection $S_X \to S_Y$ with inverse $\tau \mapsto f^{-1}\tau f$, and $f\sigma\sigma' f^{-1} = (f\sigma f^{-1})(f\sigma' f^{-1})$. □

Exercise 11.8.7

(a) The **automorphism group** $\mathrm{Aut}(G) = \{\varphi : G \to G \text{ automorphism}\}$ is a subgroup of S_G.
(b) For $g \in G$, $\mathrm{Int}(g)(x) = gxg^{-1}$ defines $\mathrm{Int}(g) \in \mathrm{Aut}(G)$, an **inner automorphism**; $\mathrm{Int}(G) = \{\mathrm{Int}(g) : g \in G\} \leq \mathrm{Aut}(G)$.
(c) (Remark) Automorphisms that are not inner are **outer**.

Solution. (a) Composites and inverses of automorphisms are automorphisms (Exercise 11.5.6), and $id \in \mathrm{Aut}(G)$. (b) $\mathrm{Int}(g)(xy) = gxyg^{-1} = gxg^{-1}gyg^{-1}$. It is bijective with inverse $\mathrm{Int}(g^{-1})$. And $\mathrm{Int}(g) \circ \mathrm{Int}(h) = \mathrm{Int}(gh)$, so $\mathrm{Int}(G)$ is closed under composition and inverses. □

Theorem 11.8.8 (Cayley)

$\lambda : G \to S_G$, $\lambda(x) = \lambda_x$ with $\lambda_x(g) = xg$, is a monomorphism. Hence every group is isomorphic to a subgroup of a symmetric group.

Proof. $\lambda_x \in S_G$ (Observation 11.2.15); $\lambda_{xy}(g) = xyg = \lambda_x\lambda_y(g)$; and $\lambda_x = id$ gives $x = \lambda_x(1) = 1$. Apply Observations 11.6.7, 11.6.6(b). ■

(Right multiplication gives $\rho_{xy} = \rho_y \circ \rho_x$, an anti-homomorphism.)

Definition 11.8.9 (Conjugacy)

$x \sim y$ iff $y = g^{-1}xg$ for some $g \in G$; this is an equivalence relation, the **conjugacy relation**, and its classes are the **conjugacy classes** (x is **conjugate** to y).

Exercise 11.8.10

(a) The conjugacy class of 1 is $\{1\}$.
(b) If G is commutative, conjugacy is equality.
(c) How many conjugacy classes does S_3 have?

Solution. (a) $g^{-1}1g = 1$.

(b) $g^{-1}xg = x$.

(c) Three: $\{id\}$, the three transpositions, the two 3-cycles (conjugation preserves cycle type, and $\sigma^{-1}\tau\sigma = \tau^2$, $\tau^{-1}\sigma\tau = \sigma\tau^2$, ...). □

Exercise 11.8.11

(a) For $H \leq G$, $g \in G$: $g^{-1}Hg = \{g^{-1}hg : h \in H\} \leq G$.
(b) On the set $\mathcal{S}$ of subgroups of G, $H \sim K$ iff $K = g^{-1}Hg$ for some g is an equivalence relation (K is a **conjugate subgroup** of H).

Solution. (a) $g^{-1}Hg = \mathrm{Int}(g^{-1})(H)$, the image of a subgroup under a homomorphism.

(b) Reflexive with $g = 1$; symmetric since $H = gKg^{-1}$; transitive since $h^{-1}(g^{-1}Hg)h = (gh)^{-1}H(gh)$. □

Exercise 11.8.12

(a) Give the Cartesian product $G = G_1 \times \cdots \times G_n$ of groups a natural group structure. (b) The i-th **projection** $\pi_i : G \to G_i$ is an epimorphism and the j-th **embedding** $\iota_j : G_j \to G$ is a monomorphism.

Solution. (a) Componentwise multiplication $(g_i)(h_i) = (g_ih_i)$; identity $(1, \ldots, 1)$, inverse (g_i^{-1}).

(b) $\pi_i(g) = g_i$ is a surjective homomorphism; $\iota_j(g) = (1, \ldots, g, \ldots, 1)$ (g in position j) is an injective homomorphism. □

Exercise 11.8.13

(a) $G \times H \approx H \times G$.
(b) $(G \times H) \times K \approx G \times (H \times K) \approx G \times H \times K$.

Solution. $(g, h) \mapsto (h, g)$ and $((g, h), k) \mapsto (g, h, k)$ and $(g, (h, k)) \mapsto (g, h, k)$ are bijective homomorphisms. □

Exercise 11.8.14

$G_1 \times \cdots \times G_n$ is commutative iff all G_i are commutative.

Solution. If all G_i commute, products commute componentwise. Conversely each $G_i \approx \iota_i(G_i)$ is a subgroup of a commutative group. □

Exercise 11.8.15

(a) $\mu_3 \times \mu_3 \not\approx \mu_9$.
(b) $\mu_2 \times \mu_3 \approx \mu_6$.

Solution. (a) Elements of $\mu_3 \times \mu_3$ have order dividing 3; μ_9 has one of order 9.

(b) $(-1, \omega)^k = 1$ ($\omega = e^{2\pi i/3}$) iff $2 \mid k$ and $3 \mid k$, so it has order 6 and $\mu_2 \times \mu_3 \approx \mu_6$ (Exercise 11.7.9). □

Exercise 11.8.16

In Exercise 11.8.12(b): (a) find $\ker \pi_i$ and $\mathrm{im}\, \iota_j$;
(b) $\pi_i \circ \iota_i = id$;
(c) $\pi_i \circ \iota_j = \mathbf{1}$ for $i \neq j$.

Solution. (a) $\ker \pi_i = \{g : g_i = 1\}$ and $\mathrm{im}\, \iota_j = \{g : g_k = 1 \text{ for } k \neq j\}$. (b), (c) are direct. □

Remark 11.8.17

The pointwise product $(\varphi\psi)(g) = \varphi(g)\psi(g)$ of homomorphisms need not be one: $\varphi(x)\varphi(y)\psi(x)\psi(y) \neq \varphi(x)\psi(x)\varphi(y)\psi(y)$ unless H is commutative.

11.9 Linear groups

Motivation. A basis identifies $\mathrm{GL}(V)$ with $\mathrm{GL}_n(F)$, and every finite group is a matrix group.

Example 11.9.1

(a) $\mathrm{GL}_n(F) = \{A \in \mathfrak{M}_{n,n}(F) : A \text{ invertible}\}$, the **general linear group**, is a group under matrix multiplication; $F^\times = \mathrm{GL}_1(F) = F - \{0\}$ is the **unit** (multiplicative) **group** of F.
(b) Subgroups of $\mathrm{GL}_n(F)$ are **linear** (**matrix**) **groups**.
(c) The **special linear group** $\mathrm{SL}_n(F) = \{A \in \mathrm{GL}_n(F) : \det A = 1\} = \ker(\det : \mathrm{GL}_n(F) \to F^\times)$ is a subgroup.

Example 11.9.2

For a finite-dimensional F-vector space V: (a) $\mathrm{GL}(V) = \{L \in \mathfrak{L}(V,V) : L \text{ invertible}\}$ is a group under composition; (b) its subgroups are **linear groups**;
(c) $\mathrm{SL}(V) = \{L \in \mathrm{GL}(V) : \det L = 1\} \leq \mathrm{GL}(V)$.

Example 11.9.3

(a) For $\dim V = n$ and a basis $\mathfrak{B}$, the vector space isomorphism $\Psi_{\mathfrak{B}}^{\mathfrak{B}} : \mathfrak{L}(V,V) \to \mathfrak{M}_{n,n}(F)$, $L \mapsto [L]_{\mathfrak{B}}^{\mathfrak{B}}$, restricts to a group isomorphism $\mathrm{GL}(V) \to \mathrm{GL}_n(F)$ ($[LM] = [L][M]$).

$$\begin{array}{ccc} \mathrm{GL}(V) & \hookrightarrow & \mathfrak{L}(V,V) \\ {\scriptstyle \Psi_{\mathfrak{B}}^{\mathfrak{B}}}\downarrow & & \downarrow{\scriptstyle \Psi_{\mathfrak{B}}^{\mathfrak{B}}} \\ \mathrm{GL}_n(F) & \hookrightarrow & \mathfrak{M}_{n,n}(F) \end{array}$$

(c) Since $\det L = \det[L]_{\mathfrak{B}}^{\mathfrak{B}}$, it also restricts to $\mathrm{SL}(V) \approx \mathrm{SL}_n(F)$.

Example 11.9.4

If $\dim V = \dim W = n$, then $\mathrm{GL}(V) \approx \mathrm{GL}(W)$. (a) Via bases: $(\Psi_{\mathfrak{C}}^{\mathfrak{C}})^{-1} \circ \Psi_{\mathfrak{B}}^{\mathfrak{B}}$. (b) Via a vector space isomorphism $\varphi : V \to W$ (e.g. with $\varphi(\mathfrak{B}) = \mathfrak{C}$): $\Gamma(L) = \varphi \circ L \circ \varphi^{-1}$, with $\Gamma(LL') = \Gamma(L)\Gamma(L')$ and $\Gamma^{-1}(M) = \varphi^{-1}M\varphi$. (c) If $\varphi(\mathfrak{B}) = \mathfrak{C}$ they coincide: $[\varphi L \varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{C}} = [\varphi]_{\mathfrak{B}}^{\mathfrak{C}}[L]_{\mathfrak{B}}^{\mathfrak{B}}[\varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{B}} = [L]_{\mathfrak{B}}^{\mathfrak{B}}$, as $[\varphi]_{\mathfrak{B}}^{\mathfrak{C}} = I$.

Example 11.9.5

$\mu_2 < \mathbb{R}^\times < \mathbb{C}^\times$ and $\mu_n < \mathbb{C}^\times$.

Example 11.9.6

Theorem 1.3.6(3) (every invertible matrix is a product of elementary matrices) says: $\mathrm{GL}_n(F)$ is generated by elementary matrices.

Example 11.9.7

$\mathbf{T}_n(F) = \{A \in \mathrm{GL}_n(F) : A \text{ upper triangular}\}$ and $\mathbf{V}_n(F) = \{A \in \mathbf{T}_n(F) : A \text{ unipotent}\}$ (diagonal entries 1) are matrix groups.

Proof [Supplement]. Products and inverses (back substitution) of invertible upper triangular matrices are upper triangular, with diagonal entries multiplied resp. inverted. ■

Exercise 11.9.8

$(F,+)$ is a matrix group: $(F,+) \approx \mathbf{V}_2(F)$.

Solution. $a \mapsto \left(\begin{smallmatrix}1 & a\\ 0 & 1\end{smallmatrix}\right)$ is a bijection, and $\left(\begin{smallmatrix}1 & a\\ 0 & 1\end{smallmatrix}\right)\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right) = \left(\begin{smallmatrix}1 & a+b\\ 0 & 1\end{smallmatrix}\right)$. □

Exercise 11.9.9

The group $(F^3, \bullet)$ of Exercise 11.1.14 is isomorphic to $\mathbf{V}_3(F)$.

Solution. $(a_1,a_2,a_3)^t \mapsto \begin{pmatrix}1 & a_1 & a_2\\ 0 & 1 & a_3\\ 0 & 0 & 1\end{pmatrix}$; the product has entries a_1+b_1, $a_2+b_2+a_1b_3$, a_3+b_3, the formula for $\bullet$. □

Example 11.9.10

Conjugacy in $\mathrm{GL}_n(F)$ is the similarity relation of §5.5.

Exercise 11.9.11

If $L \in \mathrm{GL}(V)$ is unipotent and $L \neq I$, then $|L| = \infty$ in $\mathrm{GL}(V)$ (over $F = \mathbb{R}$ or $\mathbb{C}$).

Solution. $L = I + N$, $0 \neq N$ nilpotent, $k \geq 1$ largest with $N^k \neq 0$. Then $L^m - I = \sum_{1 \leq j \leq k} \binom{m}{j} N^j$ with coefficient $m \neq 0$ at N, and $I, N, \ldots, N^k$ are independent (for $\sum c_j N^j = 0$ with first nonzero c_r, multiply by N^{k-r}: $c_r N^k = 0$). So $L^m \neq I$ for $m \geq 1$. □

Exercise 11.9.12

For fixed $C \in \mathrm{GL}_n(\mathbb{R})$, $\varphi(A) = CAC^{-1}$ and $\psi(A) = (A^t)^{-1}$ belong to $\mathrm{Aut}(\mathrm{SL}_n(\mathbb{R}))$. If $C \notin \mathrm{SL}_n(\mathbb{R})$, are φ, ψ outer automorphisms?

Solution. Both preserve $\mathrm{SL}_n(\mathbb{R})$ (det is unchanged) and products ($((AB)^t)^{-1} = (A^t)^{-1}(B^t)^{-1}$), and are bijective ($\psi^2 = id$, $\varphi^{-1} = \mathrm{Int}(C^{-1})$).

[Supplement] φ. $C' \in \mathrm{SL}_n(\mathbb{R})$ induces φ iff $C^{-1}C'$ commutes with $\mathrm{SL}_n(\mathbb{R})$ (all type-3 elementary matrices), i.e. is λI. So φ is inner iff $\lambda^n \det C = 1$ for some real λ: always for odd n; never for even n with $\det C < 0$ (outer).

ψ. For $n = 2$, $(A^t)^{-1} = JAJ^{-1}$ with $J = \left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$: inner. For $n \geq 3$, outer, since inner automorphisms preserve trace: $A = \mathrm{diag}(2, 3, \frac{1}{6}, 1, \ldots, 1)$ has trace $5\frac{1}{6} + (n-3)$, $\psi(A) = \mathrm{diag}(\frac{1}{2}, \frac{1}{3}, 6, 1, \ldots, 1)$ has trace $6\frac{5}{6} + (n-3)$. □

Exercise 11.9.13

For the invertible diagonal matrices $\mathbf{D}_n(F)$: (a) $\mathbf{D}_n(F) \approx F^\times \times \cdots \times F^\times$ (n factors); (b) $\mathbf{D}_n(F) \cap \mathrm{SL}_n(F) \approx (F^\times)^{n-1}$ for $n \geq 2$.

Solution. (a) $\mathrm{diag}(a_1, \ldots, a_n) \mapsto (a_1, \ldots, a_n)$.

(b) The last entry is determined by $a_n = (a_1 \cdots a_{n-1})^{-1}$; project to the first $n-1$ entries. □

Exercise 11.9.14

(a) If $m \leq n$, $\mathrm{GL}_m(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.
(b) If $n_1 + \cdots + n_r = n$, $\mathrm{GL}_{n_1}(F) \times \cdots \times \mathrm{GL}_{n_r}(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.

Solution. (b) $(A_1, \ldots, A_r) \mapsto A_1 \oplus \cdots \oplus A_r$ (block diagonal) is a monomorphism. (a) is the case $A \mapsto A \oplus I_{n-m}$. □

Exercise 11.9.15

(a) $\mathbf{SO}(n) < \mathbf{O}(n) < \mathrm{GL}_n(\mathbb{R})$;
(b) $\mathbf{SU}(n) < \mathbf{U}(n) < \mathrm{GL}_n(\mathbb{C})$;
(c) $\mathbf{SO}(n) = \mathrm{SL}_n(\mathbb{R}) \cap \mathbf{O}(n)$;
(d) $\mathbf{SU}(n) = \mathrm{SL}_n(\mathbb{C}) \cap \mathbf{U}(n)$.

Solution. By Exercises 9.3.3, 9.3.10, 10.6.10 and the definitions. □

Example 11.9.16

$\Psi_{\mathcal{E}}^{\mathcal{E}}$ restricts to $\mathbf{O}(\mathbb{R}^n) \approx \mathbf{O}(n)$ (dot product), and so does $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ for any orthonormal $\mathfrak{B}$ (Remark 10.6.4, Exercise 9.6.11).

Exercise 11.9.17

(a) $\mathbf{U}(\mathbb{C}^n) \approx \mathbf{U}(n)$.
(b) $\mathbf{SO}(\mathbb{R}^n) \approx \mathbf{SO}(n)$.
(c) Define $\mathbf{SU}(\mathbb{C}^n)$ and show $\mathbf{SU}(\mathbb{C}^n) \approx \mathbf{SU}(n)$.

Solution. As in Example 11.9.16, restricting $\Psi_{\mathcal{E}}^{\mathcal{E}}$; for (b), (c) also $\det L = \det[L]_{\mathcal{E}}^{\mathcal{E}}$, with $\mathbf{SU}(\mathbb{C}^n) = \{L \in \mathbf{U}(\mathbb{C}^n) : \det L = 1\}$. □

Example 11.9.18

If $\varphi : V \to W$ is an inner product space isomorphism, $\Gamma(L) = \varphi L \varphi^{-1}$ (Example 11.9.4) restricts to $\mathbf{O}(V) \approx \mathbf{O}(W)$ (resp. $\mathbf{U}$), since L preserves inner products iff $\varphi L \varphi^{-1}$ does.

Example 11.9.19

The **circle group** $S^1 = \{X \in \mathbb{R}^2 : \|X\| = 1\} = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$, and $\varphi : \mathbf{SO}(2) \to S^1$, $\varphi(R_\theta) = e^{i\theta}$, is a group isomorphism.

Exercise 11.9.20

$\mathbf{SO}(2) \approx S^1 \approx \mathbf{U}(1)$.

Solution. $\varphi(R_\theta R_\eta) = \varphi(R_{\theta+\eta}) = e^{i(\theta+\eta)} = e^{i\theta}e^{i\eta}$, and φ is bijective (Theorem 9.5.2). $\mathbf{U}(1) = \{z \in \mathbb{C} : \bar{z}z = 1\} = S^1$. □

Exercise 11.9.21

$S_n \approx \{I_\sigma \in \mathfrak{M}_{n,n}(F) : \sigma \in S_n\} \approx \{P_\sigma \in \mathfrak{L}(V,V) : \sigma \in S_n\}$ ($\dim V = n$), via permutation matrices and permutation operators.

Solution. The permutation matrix I_σ (with $I_\sigma e_j = e_{\sigma(j)}$) satisfies $I_\sigma I_\tau = I_{\sigma\tau}$, and $\sigma \mapsto I_\sigma$ is injective; similarly for P_σ on a basis. □

Corollary 11.9.22

Every finite group is isomorphic to a linear (matrix) group.

Proof. By Cayley's theorem G embeds in $S_G \approx S_n$ ($n = |G|$), and S_n embeds in $\mathrm{GL}_n(F)$ by Exercise 11.9.21. ■

Example 11.9.23

In $\mathbf{O}(2)$ let $R = R_{2\pi/3}$, $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $D_3 = \langle R, S\rangle \leq \mathbf{O}(2)$.
(a) The plane rule gives $RS = SR^2$. D_3 consists of words in R, R^2, S; moving each S left gives $D_3 = \{S^iR^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\}$, six distinct matrices (determinants, first columns). $S^2 = I$, $R^3 = I$, $RS = SR^2$ determine the table; e.g. $(SR)(SR^2) = S(RS)R^2 = S^2R^4 = R$.
(b) These are the relations $\sigma^2 = 1$, $\tau^3 = 1$, $\tau\sigma = \sigma\tau^2$ of S_3 (Example 11.8.4).
(c) So $\sigma^i\tau^j \mapsto S^iR^j$ is a table-preserving bijection: $S_3 \approx D_3$.

Exercise 11.9.24

For $n \geq 3$, the **dihedral group** $D_n = \langle R_{2\pi/n}, S\rangle \leq \mathbf{O}(2)$ has $|D_n| = 2n$.

Solution. $R^n = I$, $S^2 = I$, $RS = SR^{-1}$ give $D_n = \{S^iR^j : 0 \leq i \leq 1, 0 \leq j < n\}$ as in Example 11.9.23; distinct since $\det(S^iR^j) = (-1)^i$ and the R^j are distinct. □

Exercise 11.9.25

(a) For $\sigma = (2\,5)(3\,4)$, $\tau = (1\,2\,3\,4\,5) \in S_5$, $G_5 = \langle\sigma,\tau\rangle \approx D_5$. (b) For $\sigma = (2\,6)(3\,5)$, $\tau = (1\,2\,3\,4\,5\,6) \in S_6$, $G_6 = \langle\sigma,\tau\rangle \approx D_6$. (c) Interpret (a),
(b) as permutations of the vertices of a regular pentagon/hexagon, and generalize to the regular n-gon.

Solution. $\sigma^2 = 1$, $\tau^n = 1$, $\sigma\tau\sigma = \tau^{-1}$ (e.g. $\sigma\tau\sigma(1) = \sigma(2) = n = \tau^{-1}(1)$), so $\sigma^i\tau^j \mapsto S^iR^j$ is an isomorphism onto D_n (Example 11.9.23). With vertices of the regular n-gon numbered counterclockwise, τ is rotation by $2\pi/n$ and σ the reflection through vertex 1: D_n is the symmetry group of the n-gon. □